

YunTAPR-Net B0 Phase-B FinalFit

Formal Training Completion and Evidence Report v1

Run: run_20261002T035929_487644Z

Baseline: 4de37ac087183bddf3b9f8c7d548a0f976a50e91

Scope: B0 Phase-B FinalFit only.

Completed authorized run

All eleven fixed-budget epochs completed, with 23,447 frozen identities exactly once in each epoch. The training population contains 11,720 scenes from 2023 and 11,727 from 2024. No 2025 scenes or pixels were read. Each epoch contains 11,723 full batches of two scenes and one actual singleton: 11,724 optimizer updates, 128,964 in total. No identities were skipped, duplicated, padded, replaced or dropped. The singleton denominator is 3,430; a full batch uses 6,860.

Initialization used fresh seed 2026. Its logical model state SHA256 matched the independently verified fresh initialization identity before training:

57a4d103a31aa7be1a52af079cdf7fb81bc73c51ba3e0e97d395513d21d9023d

No Phase-A model, optimizer or scheduler state initialized FinalFit. Historical authorization fields and all 1,656 baseline files remained unchanged.

Unchanged scientific and numerical protocol

The frozen Phase-B normalization was reused, without refitting:

$$\mu = 270.5900486586461 \text{ K}, \quad \sigma = 20.368583874067266 \text{ K}.$$

Artifact SHA256:

c7042beac2412594ca9fc264d7df10981755b393158c539cb48c9b56cd314327

Manifest SHA256:

00e6bd018eae6aeb05be0740b55a92b7a5cc9a44b9ab2b80affbad3e3282ccff

AdamW used $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$ and weight decay 10^{-4} for Conv2d kernels only. Forward precision was BF16; parameters and raw quantiles were FP32. Quantile transformations and pinball calculations used FP64. No GradScaler was used. Global gradient clipping remained 5, with nonfinite errors causing STOP. Architecture, loss and hyperparameters were unchanged.

The stateless scheduler retained the frozen 50-epoch horizon: $W = 11,724$, $U = 586,200$, base LR 10^{-4} and cosine minimum 10^{-6} . The actual first-epoch singleton LR was exactly 10^{-4} . The cosine horizon was not compressed to eleven epochs.

There was no independent 2024 validation, early stopping, BEST selection, epoch reselection, threshold calibration or training-loss budget adjustment.

Actual epoch evidence

The losses below are global raw numerator sums divided by the total actual valid-pixel denominator, not averages of batch losses. Each epoch contains 80,423,210 valid pixels. Full precision values and all requested I/O, rainy-pixel, LR, clipping, GPU and singleton metrics are preserved in `training_history.csv` and the eleven epoch summary JSON files.

Epoch	Scenes	Updates	Occurrence	Quantile	Core	End LR
1	23,447	11,724	0.03448658	0.03858595	0.07307252	0.0001
2	23,447	11,724	0.03200169	0.01448328	0.04648497	9.98983e-05
3	23,447	11,724	0.03099689	0.0141999	0.04519679	9.959361e-05
4	23,447	11,724	0.03028817	0.01400176	0.04428993	9.908718e-05
5	23,447	11,724	0.02965103	0.01383817	0.04348921	9.83811e-05
6	23,447	11,724	0.02919559	0.01370123	0.04289682	9.747826e-05
7	23,447	11,724	0.02875113	0.01357739	0.04232852	9.638238e-05
8	23,447	11,724	0.02835876	0.01346868	0.04182744	9.509796e-05
9	23,447	11,724	0.02801436	0.01334646	0.04136082	9.363028e-05
10	23,447	11,724	0.02764786	0.01322562	0.04087348	9.198536e-05
11	23,447	11,724	0.02736705	0.01311352	0.04048058	9.016997e-05

Epoch	Norm min	Norm max	Norm mean	Clips	GPU GiB	Minutes
1	0.010125	23.596	0.896	89	2.670	50.24
2	0.0036288	4.3711	0.33657	0	2.670	50.11
3	0.0030602	2.3496	0.23979	0	2.670	50.08
4	0.0043125	2.3836	0.19901	0	2.670	50.10
5	0.0033622	1.5836	0.17405	0	2.670	50.07
6	0.0042983	1.4916	0.1625	0	2.670	50.12
7	0.0033529	1.5495	0.15112	0	2.670	50.91
8	0.0026405	1.131	0.1435	0	2.670	50.68
9	0.0030733	1.5721	0.13591	0	2.670	50.54
10	0.004444	1.1662	0.13054	0	2.670	50.07
11	0.0027811	1.138	0.12741	0	2.670	50.14

The GPU column reports peak allocated memory. Peak reserved memory, summed source staging/read times, singleton identity and its exact LR remain in the CSV/JSON evidence. Displayed numbers are rounded only for this PDF.

Checkpoint identity and integrity

Only completed epoch boundaries produced checkpoint payloads. LAST and FINAL both identify epoch 11, global update 128,964. All eleven boundary files remain under the approved local F: root. Temporary write, fsync/close, round-trip verification, SHA checks and atomic rename were performed. Retained boundary payloads were independently reverified after training.

FINAL local path:

```
F:\pytorch\Research\outputs\formal_training\b0_phase_b_finalfit/  
run_20261002T035929_487644Z\epoch_011.pt
```

FINAL SHA256:

```
05359d2fee2ae61daf654ae5a59b7977cb65247fd46d97a69a690a0133da7f65
```

Binary model and optimizer states are excluded from Git. Checkpoint registry and identity JSON preserve paths, sizes, hashes, epoch/update counters, authorization and complete environment/provenance metadata.

Post-training closure

The actual local update log was reread to check every sample identity, seeded order, LR, precision gate, denominator and optimizer update. The eleven epochs were independently checked for exact coverage. Global losses, gradient statistics, clipping and singleton metadata were recomputed and compared with the JSON and CSV histories. FINAL passed read-only readability, SHA256 and provenance verification.

The current 242 regression tests and 12 legal new formal-run tests were rerun after training: 254 actual tests, zero failures, errors or skips. Tests used isolated fixtures, cleaned all temporary test artifacts and never applied FINAL state to a test model. FINAL SHA256 before and after tests was identical. Original source identity and runtime read/QC/causality checks passed; no source or numerical failure was bypassed.

Phase-B authorized	true
Formal training started	true
FinalFit completed	true
Completed epochs	11
Formal optimizer updates	128,964
FINAL epoch	11
2025 pixels read	0
B1 started / B1–B8 authorized	false / false
2025 Final Test authorized / executed	false / false

Evidence and limits

The run directory contains independent authorization references, `formal_run_manifest.json`, `training_history.csv`, eleven summaries and histories, checkpoint registry and identities, runtime and verification reports, actual post-training test logs, final status and artifact hashes. This editable LaTeX source accompanies the PDF. The machine-readable evidence contains full precision numbers and exact provenance.

A report-only execution-manifest path error after all tests passed was recorded. Its failure history and timestamped metadata correction are preserved. The correction rechecked FINAL SHA/provenance and added zero formal optimizer updates.

This report establishes fixed-budget FinalFit completion. It makes no 2025 Final Test or B1–B8 performance claim. Execution stops after evidence publication. Subsequent stages remain unauthorized.